Nida Esen

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Education

Northeastern University — San Jose, CA, USA
Master of Science in Artificial Intelligence — *Sep. 2025 – Expected Dec. 2027*
- **Relevant Coursework:** Algorithms, Artificial Intelligence, Machine Learning, NLP, Programming Design Paradigm (PDP)

Uludag University — Bursa, Turkiye
Bachelor of Science in Computer Engineering — *Sep. 2019 – Aug. 2023*

Technical Skills

Concepts: Imitation Learning (ACT), Vision-Language-Action Models ($\pi_{0.5}$), Teleoperation Data Collection, Sim-to-Real, Sensor Fusion, RL (MAPPO, Constrained RL)
Robotics & Simulation: ROS2, LeRobot, Isaac Sim, MuJoCo, Gazebo, PettingZoo, NVIDIA Jetson
ML & Computer Vision: PyTorch, TensorFlow, Hugging Face, Scikit-learn, Numba, OpenCV, YOLOv8, CLIP, TensorRT
Languages: Python, C++, SQL, Java, TypeScript, JavaScript, PHP
Cloud & MLOps: AWS (EC2, S3, Lambda, Bedrock, CloudWatch, IAM), Docker, Git, Linux, CUDA, SLURM, MLflow, W&B
LLMs & Retrieval: LangChain, LangGraph, Gemini, LLaMA, Sentence Transformers, Qdrant, ChromaDB, FAISS

Experience

Northeastern University, Physical Hardware Intelligence Lab — San Jose, CA
AI/ML Developer — *Sep. 2026 – Present*
- **Phi Studio:** Built a local app for the full robot-learning loop (setup, teleoperation with a live 3D twin, LeRobot dataset recording, blind evaluation); auditing 120 teleop episodes surfaced a 108 ms leader–follower lag and 68 idle starts.
- **ACT Imitation Learning:** Trained an Action Chunking with Transformers (ACT) policy from teleoperated demonstrations for cube pick-and-place; currently extending it to precision battery insertion.
- **Sim-to-Real RL with Squint:** Deployed a sim-trained visual RL policy on SO-101 arms, aligning the simulated camera pose to the real wrist mount (view IoU 0.41 vs. $\geq$0.66 in training) and auditing 55 sim-to-real gaps; 5/9 real-arm PlaceCube successes (5/5 on close layouts) at 15 Hz on CPU.
- **Two-Arm Laundry Folding:** Reproducing an ICRA 2026 LeHome prize-winning $\pi_{0.5}$ VLA (AWR + RECAP RL) on a new dual-arm rig, toward an end-to-end basket-to-fold pipeline for kids' garments; 6/8 zero-shot real-robot towel folds, 13/16 unseen garments in Isaac Sim, and 59% lower latency (0.68 s $\rightarrow$ 0.28 s).

Uludag University, Autonomous Car Research Group — Bursa, Turkiye
Computer Vision Undergraduate Research Assistant — *2021 – Aug. 2023*
- Built a real-time YOLOv8 detection and segmentation perception stack for a Renault-supported autonomous vehicle, reaching 92% mAP@0.5 at 30 FPS on NVIDIA Jetson and cutting inference latency 40% through TensorRT optimization.
- Designed a camera–LiDAR late-fusion pipeline as modular ROS2 nodes, fusing YOLOv8 and LiDAR detections at the object level; cut false positives 25% vs. camera-only on the KITTI 3D benchmark, with <50 ms latency in Gazebo.

Renault Group, Advanced Driver Assistance Systems — Bursa, Turkiye
AI Engineer Intern — *Oct. 2022 – Mar. 2023*
- Co-engineered a camera-based driver drowsiness monitoring system shipped in the Renault Clio (219K+ units sold in Europe, 2023); built the eye-blink and attention detection model achieving 93% recall at <1 false alert/hour with 30 ms latency on the in-vehicle ECU.
- Led Intelligent Speed Assist (ISA) regulatory analysis for EU/Turkiye, informing L2+/L3 feature strategy.

Renault Group, Information Technologies — Bursa, Turkiye
Software Engineer Intern — *Mar. 2023 – Jun. 2023*
- Built and deployed an ISO 27001–compliant, scalable full-stack web application (Laravel, MySQL, JavaScript, PHP) reactivating Renault's production quality system for 200+ staff across cross-functional engineering teams.

Research Experience

Northeastern University, Graduate Thesis Research — San Jose, CA
Dose-Budgeted Multi-Agent Reinforcement Learning for Radioactive Source Search — *Sep. 2026 – Present*
- Developing constrained multi-agent RL (MAPPO with per-agent Lagrangian safety constraints, PyTorch) for dose-budgeted radioactive source search: robot teams that locate a source while keeping each robot within its radiation exposure limit.
- Built a physics-based simulator (PettingZoo, Numba) with particle-filter estimation, a MuJoCo 3D motion layer for sim-to-sim tests, Gazebo world export, and a Hydra/SLURM evaluation pipeline.

Projects

Odin's Eye: Crowd Tracking + VLM Narration | *PyTorch, ByteTrack, DINOv2, Qwen3-VL* — Nov. 2026
- Researching dense multi-object tracking on MOT20 ($\sim$150 pedestrians/frame): extending a YOLO26m + ByteTrack + DINOv2 ReID tracker (baseline 71% MOTA) and its Qwen3-VL scene-narration layer to cut identity switches under heavy occlusion.

RoboScript AI: Edge LLM for AMR Fleet Control | *Pydantic, Hugging Face, TRL (DPO), ROS 2* — Jun. 2026
- Building an LLM that maps warehouse-robot telemetry (LiDAR, YOLO detections, fault codes) to one schema-validated ROS 2 action across 12 fault types; built its synthetic-data pipeline (Pydantic validation, LLM-as-judge, DPO pairs).